

DATA 581

Modeling and Simulation II



The Markov Property and Linear Time Series Models

Outline

1. Markov Property
2. Linear Time Series: Autoregressive Time Series as Markov processes
3. Non-Markovian Linear Time Series: Moving Average, ARMA and ARIMA

```
set.seed(208721)
```

The Markov Property

Recall that if a sequence of random variables $Z_1, Z_2, \dots, Z_n$ is independent, then their joint distribution can be factored:

$$f(z_1, z_2, \dots, z_n) = f(z_1)f(z_2) \cdots f(z_n).$$

If the sequence is completely dependent, the factorization involves complicated conditional distributions along the lines of:

$$\begin{aligned} f(z_1, z_2, \dots, z_n) &= f(z_n | z_1, z_2, \dots, z_{n-1}) f(z_{n-1} | z_1, \dots, z_{n-2}) \cdots \\ &\cdots f(z_4 | z_1, z_2, z_3) f(z_3 | z_1, z_2) f(z_2 | z_1) f(z_1). \end{aligned}$$

Back in the 19th century, A. Markov realized that there might be a useful class of models in between these two extremes.

The Markov Property

$f(z_3|z_1, z_2)$ can be approximated by $f(z_3|z_2)$. This will usually be a better approximation than $f(z_3)$ only - which is what independence implies. This approximation says that z_3 is dependent on z_1 only through z_2 explicitly. In other words, given z_2 , no additional information in the sample will provide information about z_3 .

Similarly, approximate $f(z_4|z_1, z_2, z_3)$ by the simpler $f(z_4|z_3)$, and so on, finally approximating $f(z_n|z_1, \dots, z_{n-1})$ by $f(z_n|z_{n-1})$.

That is, assume

$$f(z_1, z_2, \dots, z_n) = f(z_n|z_{n-1})f(z_{n-1}|z_{n-2}) \cdots \\ \cdots f(z_4|z_3)f(z_3|z_2)f(z_2|z_1)f(z_1).$$

... This is the Markov property.

Autoregressive Time Series

Recall the autoregressive process of order 1:

$$Z_n = \phi Z_{n-1} + \varepsilon_n$$

where the ε 's are independent random variables with mean 0.

The sequence of Z 's is a discrete time Markov process with a *continuous* state space, since knowledge of Z_{n-2} and earlier provides no information about Z_n if Z_{n-1} is known.

AR(1) as a Markov Process

We can formally check for the Markov property:

Letting $A_n, B_n, \dots$ denote sets of points on the real lines, e.g. intervals, etc, we have

$$\begin{aligned} P(Z_n \in A_n | Z_{n-1} \in B_n, Z_{n-2} \in C_n, \dots) \\ &= P(\phi Z_{n-1} + \varepsilon_n \in A_n | Z_{n-1} \in B_n, Z_{n-2} \in C_n, \dots) \\ &= P(\phi Z_{n-1} + \varepsilon_n \in A_n | Z_{n-1} \in B_n) \\ &= P(Z_n \in A_n | Z_{n-1} \in B_n) \end{aligned}$$

because ε_n is independent of all Z 's prior to time n .

Autoregressive Time Series

Higher order autoregressive processes are also Markov processes. .

e.g. AR(2):

$$Z_n = \phi_1 Z_{n-1} + \phi_2 Z_{n-2} + \varepsilon_n$$

$\{(Z_{n-1}, Z_n)\}$ is a ***second order*** Markov process: Knowledge of any values prior to the pair of values (Z_{n-2}, Z_{n-1}) does not give us additional information about (Z_{n-1}, Z_n) .

AR(2) as a Markov Process

We can formally check for the Markov property for the successive pairs (Z_n, Z_{n-1}) :

$$\begin{aligned}
 &P(Z_n \in A_n, Z_{n-1} \in B_n | Z_{n-1} \in B'_n, Z_{n-2} \in C_n, Z_{n-3} \in D_n, \dots) \\
 &= P(\phi_1 Z_{n-1} + \phi_2 Z_{n-2} + \varepsilon_n \in A_n, Z_{n-1} \in B_n | \\
 &\quad Z_{n-1} \in B'_n, Z_{n-2} \in C_n, Z_{n-3} \in D_n, \dots) \\
 &= P(\phi_1 Z_{n-1} + \phi_2 Z_{n-2} + \varepsilon_n \in A_n, Z_{n-1} \in B_n | Z_{n-1} \in B'_n, Z_{n-2} \in C_n) \\
 &= P(Z_n \in A_n, Z_{n-1} \in B_n | Z_{n-1} \in B'_n, Z_{n-2} \in C_n) \quad .
 \end{aligned}$$

The second equality follows because ε_n is independent of $Z_{n-1}, Z_{n-2}, Z_{n-3}, \dots$

AR(p) as a Markov Process

$$Z_n = \phi_1 Z_{n-1} + \cdots + \phi_p Z_{n-p} + \varepsilon_n$$

This is a p th order Markov process: Knowledge of any values prior to Z_{n-p+1} does not give us additional information about $(Z_{n-p+1}, \dots, Z_n)$.

Exercise: show that for $p = 3$, (Z_n, Z_{n-1}, Z_{n-2}) satisfies the Markov property.

Implications of the Markov Property for Modelling and Simulation

Why worry about the technicalities of the Markov property?

1. As mentioned before, as a modelling device the Markov property presents an intermediate point between the simplest, and often unrealistic, independence assumption and the most difficult, but more realistic, but usually intractable, assumption, of full dependence.
2. When the Markov property is assumed, simulation algorithms can be constructed more easily. A Markovian time series, such as from an autoregressive process, can be simulated in time order, using a simple for loop, from an initial starting point or set of starting points (in the case of a higher order AR process).

For example, for an AR(3) process, if we know Z_1 , Z_2 and Z_3 , we can simulate a value of Z_4 from the model, and then Z_5 can be simulated using the current values of Z_2 , Z_3 and Z_4 . We don't need to keep track of all of the information or use information that might not be available.

We next consider time series models where this last statement is not true.

Moving Average Processes

Unlike AR processes where the data points are all dependent on each other, the data points of an MA process only depend on each other if they close together in time.

This offers new modelling possibilities.

The Moving Average Process of Order 1

The 0-mean moving average process of order 1 (MA(1)) is defined as

$$Z_n = \theta_1 \varepsilon_{n-1} + \varepsilon_n$$

where the ε 's are independent and normally distributed with mean 0 and variance σ^2 , and θ_1 is a constant.

According to this definition, any ε_k will be independent of Z_j for all $j < k$.

If we define $Y_n = Z_n + \mu$, then Y_n is a MA(1) process with mean μ .

Because the ε 's have expectation 0, it follows that $E[Z_n] = 0$.

Is this a Markov process?

The MA(1) process is *not* a Markov process, since

$$Z_n = \varepsilon_n + \theta_1 \varepsilon_{n-1}$$

and

$$\varepsilon_{n-1} = Z_{n-1} - \theta_1 \varepsilon_{n-2}$$

so

$$Z_n = \varepsilon_n + \theta_1 Z_{n-1} - \theta_1^2 \varepsilon_{n-2}.$$

But

$$\varepsilon_{n-2} = Z_{n-2} - \theta_1 \varepsilon_{n-3}$$

so

$$Z_n = \varepsilon_n + \theta_1 Z_{n-1} - \theta_1^2 Z_{n-2} + \theta_1^3 \varepsilon_{n-3}.$$

And continuing, we would see that Z_n depends explicitly on all previous Z 's. This means that prediction of Z_n , given Z_{n-1} could be improved upon by making use of earlier Z 's, contradicting the Markov property.

Implications for Modelling and Simulation

Since the MA(1) process is not Markovian, when we simulate it, we cannot just keep track of the Z values.

We need to keep track of the innovations - the ε 's, as well as the Z 's.

Let's simulate 10 observations from a normally distributed 0-mean MA(1) process with parameter $\theta = .5$ and an innovation standard deviation of 2.5. Since

$$Z_1 = 0.5\varepsilon_0 + \varepsilon_1$$

we immediately see that we need a previous innovation value, ε_0 . When we model real data, this will be an issue, since we do not observe Z_0 , let alone ε_0 .

Implications for Modelling and Simulation

When we simulate, it is okay, since we can just draw from a normal distribution with standard deviation 2.5:

```
eps0 <- rnorm(1, sd = 2.5)
```

We can then proceed with the rest of the simulation as follows:

```
n <- 10; eps <- eps0; Z <- numeric(n)
for (i in 1:n) {
  epsn <- rnorm(1, sd = 2.5)
  Z[i] <- 0.5*eps + epsn
  eps <- epsn
}
Z
## [1] -3.149762779 -1.068456395 -0.228774740 -3.748692269 -2.745145796
## [6] 0.179918827 2.842479061 0.719998874 0.413697957 0.001022298
```

The function `arima.sim()` can be used to simulate MA processes more efficiently.

Development of the Variance for the MA(1) Process

By squaring both sides of the defining equation and taking expectations, we have

$$E[Z_n^2] = \sigma^2(1 + \theta_1^2).$$

This is the variance of Z_n , because $E[Z_n] = 0$.

Details: Squaring the right hand side gives $\theta_1^2 \varepsilon_{n-1}^2 + \varepsilon_n^2 + 2\theta_1 \varepsilon_n \varepsilon_{n-1}$. Since the ε 's are independent, $E[\varepsilon_n \varepsilon_{n-1}] = E[\varepsilon_n]E[\varepsilon_{n-1}] = 0$ so the expected value of the right hand side is $\sigma^2 \theta_1^2 + \sigma^2$.

Development of the First Lag Autocovariance

By multiplying the defining equation by Z_{n-1} and taking expectations, show that

$$E[Z_n Z_{n-1}] = \theta_1 \sigma^2.$$

Details: To do this, you will need to use the fact that the defining equation also implies that

$$Z_{n-1} = \theta_1 \varepsilon_{n-2} + \varepsilon_{n-1}.$$

Then

$$Z_n Z_{n-1} = \varepsilon_n \varepsilon_{n-1} + \theta_1 \varepsilon_{n-1}^2 + \theta_1 \varepsilon_n \varepsilon_{n-2} + \theta_1^2 \varepsilon_{n-1} \varepsilon_{n-2}.$$

Taking expectations of this and using independence of the ε 's gives

$$E[Z_n Z_{n-1}] = \theta_1 E[\varepsilon_{n-1}^2] = \theta_1 \sigma^2.$$

Thus, the covariance of Z_n and Z_{n-1} is $\theta_1 \sigma^2$. This is the first lag autocovariance.

Development of the First Lag Autocovariance

Since $E[Z_n] = 0$, the covariance is just $E[Z_n Z_{n-1}]$.

Use the same procedure as in the preceding part to show that

$$E[Z_n Z_{n-k}] = 0$$

for $k = 2, 3, \dots$. We deduce that the covariance of all autocovariances at lags larger than 1 must be 0.

Details:

$$E[Z_n Z_{n-k}] = E[(\varepsilon_n + \theta_1 \varepsilon_{n-1})(\varepsilon_{n-k} + \theta_1 \varepsilon_{n-k-1})] = 0$$

because all of the products inside the expectation involve different ε 's and, therefore, must have expectation 0, because of independence.

Development of the First Lag Autocorrelation

The k th autocorrelation is defined as the k th autocovariance divided by the variance of the process. We deduce that all autocorrelations beyond lag 1 are 0, and that the first one is nonzero.

Details: Since the autocovariances beyond lag 1 are all 0, dividing by anything still gives 0. The lag 1 autocovariance is nonzero so dividing by anything still gives a nonzero result.

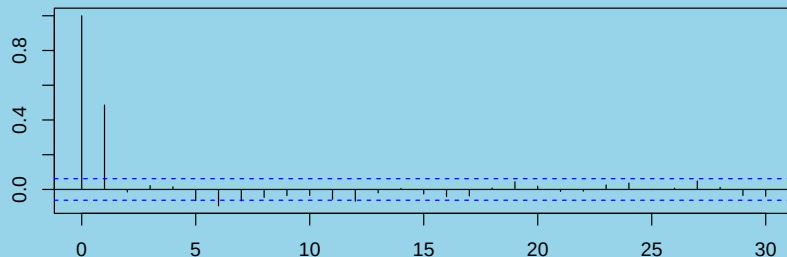
Simulation of MA(1) Data

Simulate 1000 observations from an MA(1) process with $\theta_1 = 0.9$, using the following code

```
ma1 <- arima.sim(1000, model=list(ma=c(.9)))
```

The sample autocorrelation function for the data is obtained as follows:

```
acf(ma1)
```



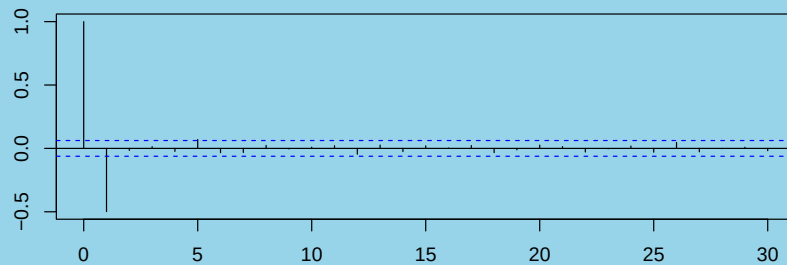
The autocorrelations are all 0 beyond lag 1 as expected, and the first one is positive which agrees with the formula.

Simulating an MA(1) Process with a Negative Parameter

Repeating the preceding simulation but with $\theta_1 = -0.9$:

```
ma1 <- arima.sim(1000, model=list(ma=c(-.9)))
```

```
acf(ma1)
```



The autocorrelations are all 0 beyond lag 1 as expected and the first one is negative which agrees with the formula.

Distinguishing Between MA(1) and AR(1) Processes

Given a set of time series data where you are faced with the question as to whether an AR(1) or MA(1) process is appropriate as a model, what action would you take, and how would you make your choice?

Answer: Plot the sample acf function and check to see if the autocorrelations cut off after lag 1 (indicating MA(1)) or if they decay exponentially (indicating AR(1)).

The Moving Average Process of Order 2 (MA(2))

This is defined as

$$Z_n = \theta_2 \varepsilon_{n-2} + \theta_1 \varepsilon_{n-1} + \varepsilon_n$$

where the ε 's are independent and normally distributed with mean 0 and variance σ^2 , θ_2 and θ_1 are constant.

According to this definition, any ε_k will be independent of Z_j for all $j < k$.

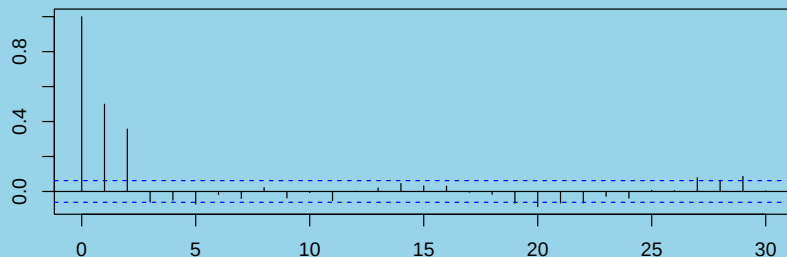
If we define $Y_n = Z_n + \mu$, then Y_n is a MA(2) process with mean μ .

Simulation of MA(2) Data

We can simulate 1000 observations from an MA(2) process with $\theta_1 = 0.5$ and $\theta_2 = 0.7$ using the following code

```
ma2 <- arima.sim(1000, model=list(ma=c(.5, .7)))
```

```
acf(ma2)
```



The autocorrelations are all 0 beyond lag 2 as expected and the first two are nonzero.

The ACF gives us a way to distinguish data following an MA(2) process from an MA(1) process.

Autoregressive-moving average process of order 1, 1 (ARMA(1, 1))

This is defined as

$$Z_n = \phi_1 Z_{n-1} + \theta_1 \varepsilon_{n-1} + \varepsilon_n$$

where the ε 's are independent and normally distributed with mean 0 and variance σ_2 , θ_2 and θ_1 are constant.

According to this definition, any ε_k will be independent of Z_j for all $j < k$.

If we define $Y_n = Z_n + \mu$, then Y_n is an ARMA(1, 1) process with mean μ .

Simulating ARMA(p,q) Data

Simulate 1000 observations from an ARMA(1, 1) process with $\phi_1 = 0.5$ and $\theta_2 = 0.7$ using the following code

```
arma11 <- arima.sim(1000, model=list(ar=c(.5), ma=c(.7)))
```

The more general ARMA(p, q) process has a pth order AR component and a qth order MA component. For example, the ARMA(2,2) process (with mean 0) is given by

$$Z_n = \phi_1 Z_{n-1} + \phi_2 Z_{n-2} + \theta_1 \varepsilon_{n-2} + \theta_2 \varepsilon_{n-1} + \varepsilon_n.$$

We simulate 1000 observations from this process (with $\phi_1 = .2, \phi_2 = -.7, \theta_1 = -.5, \theta_2 = .3$, using

```
arma22 <- arima.sim(1000, model=
  list(ar=c(.2, -.7), ma=c(-.5, .3)))
```

The Integrated Autoregressive-Moving Average Process (ARIMA)

ARIMA processes allow for random trends.

The ARIMA(1, 1, 1) is defined as

$$X_n = X_{n-1} + Z_n$$

where

$$Z_n = \phi_1 Z_{n-1} + \theta_1 \varepsilon_{n-1} + \varepsilon_n$$

and where the ε 's are independent and normally distributed with mean 0 and variance σ_2 , θ_2 and θ_1 are constant.

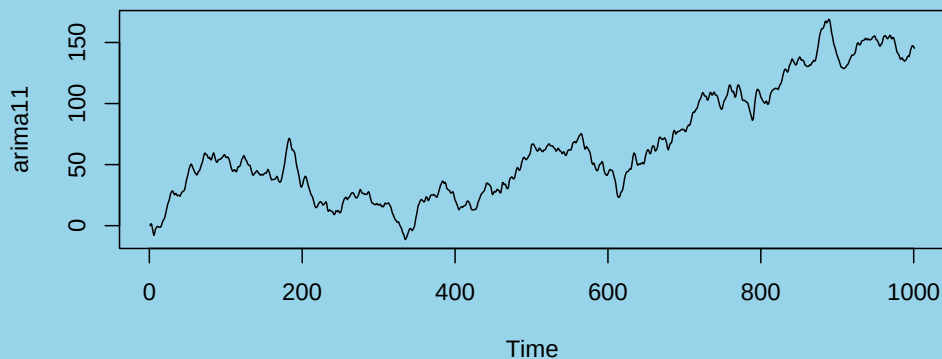
An ARIMA(1,1,1) process with drift μ is defined as $X_n = X_{n-1} + Y_n$ where $Y_n = Z_n + \mu$; this incorporates a deterministic trend as well as a random trend.

Simulating ARIMA Data

We can simulate 1000 observations from an ARMA(1, 1, 1) process with $\phi_1 = 0.5$ and $\theta_2 = 0.7$ using the following code

```
arima11 <- arima.sim(1000, model=list(order=c(1, 1, 1),
  ar=c(.5), ma=c(.7)))
```

```
ts.plot(arima11)
```



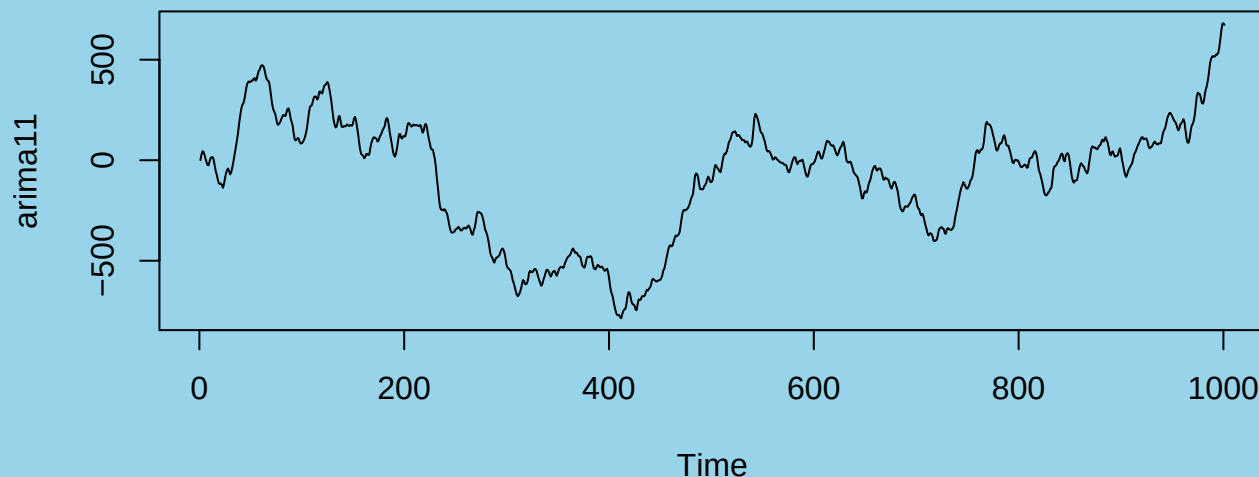
Note the apparent trend. This is not systematic and could just as likely trend downward.

Changing the Noise Standard Deviation

In order to simulate data with a different noise standard deviation, use the `sd` argument in the `arima.sim()` function as, for example, with $\sigma = 10$:

```
arima11 <- arima.sim(1000, model=list(order=c(1, 1, 1),  
    ar=c(.5), ma=c(.7)), sd=10)
```

```
ts.plot(arima11)
```

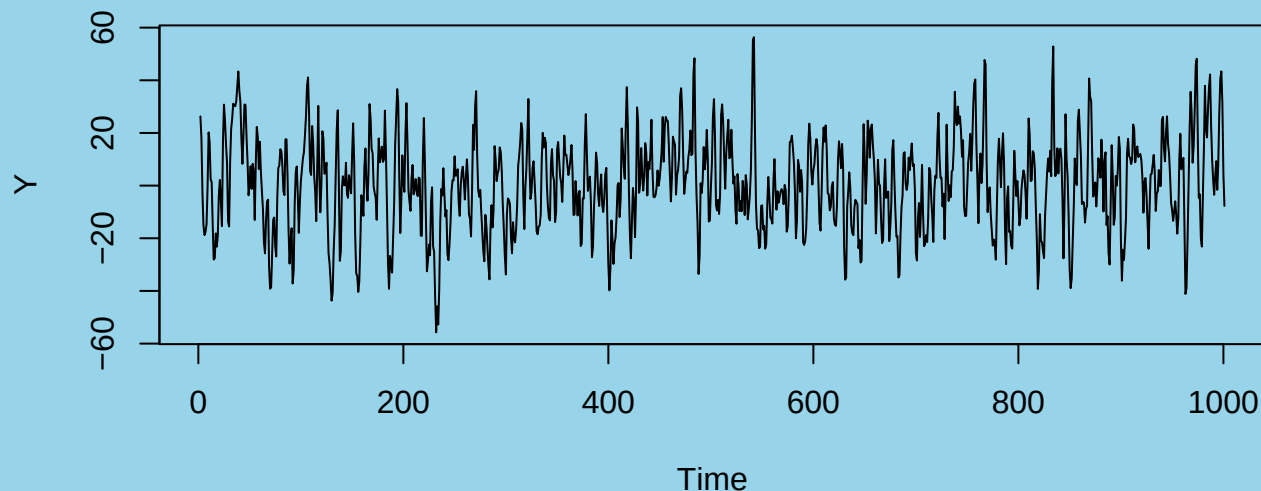


Removing a Trend by Differencing

The random trends in $\text{ARIMA}(p, 1, q)$ models can be removed by successive differencing. That is, if Z_n is $\text{ARIMA}(p, 1, q)$, then $Y_n = Z_n - Z_{n-1}$ will be a 0-mean $\text{ARMA}(p, q)$ process.

The `diff()` function can perform the successive differencing:

```
Y <- diff(arima11)
ts.plot(Y)
```



Summary

- 1. The Markov property can be used to model data that transition from discrete state to discrete state, and in both discrete and continuous time.**
- 2. In this lecture, we have seen that some time series, which have continuous states, can be modelled with the Markov property. Other time series models also exist, namely, the Moving Average and ARMA processes.**
- 3. Nonstationarity can be introduced into time series models with the integration operator - which is the opposite of successive differencing (which can remove nonstationarity).**
- 4. Simulating time series models is straightforward in the Markov case and can be implemented in non-Markov cases, but additional information must be retained in order to simulate forward in time.**

Coming Up ...

- In the next lecture, we will see that there are Markov processes that are continuous in both state and time.
- These processes include the Gaussian processes and the Wiener process, used in financial modelling.
- We will also briefly look at some of the nonlinear time series models that have become important in the modelling of financial data.